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Review of Concept Drift Detection Method for Industrial Process Modeling

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Abstract: With the advent of big data era in industry, data-driven modeling methods have been applied widely. The concept drift problem in industrial process modeling has also attracted widespread attention. However, the current research on concept drift focuses on classification tasks and computer fields, with less work on regression modeling of industrial processes. Aiming at the above problems, this paper summarizes the concept drift detection methods for industrial process modeling, and guide for solving this problem. First, the general definition of concept drift and its existence in industrial processes are introduced. Then, the existing drift detection technologies based on process variables, based on prediction error of difficulty-to-measure parameter and based on combine multiple factors are addressed. Thirdly, these methods are discussed, and some research difficulties are given out. Finally, the conclusion and the future research directions for the existing concept detection difficulties are presented.

Key Words: Industrial process, Concept drift detection, Process variable, Prediction error, Sample distribution

1 Introduction

With the increasing requirements of complex industrial processes for control, operating efficiency and reliability, it is difficult to obtain the changes of the difficulty-to-measure parameter (i.e., production quality and quantity, pollution emission index). Most of them are obtained by traditional decision-making based on expert knowledge and off-line laboratory analysis. For large and complex industrial processes, real-time difficulty-to-measure parameter information needs to be integrated to achieve effective process control. Therefore, industrial process modeling has become an important part of the system characteristics analysis, controller design, and operating optimization. However, industrial processes usually have complex structures, which need many on-site interferences. Thus, it is difficult to construct the first principal model for different research objectives. To satisfy the high requirements of security and stability, data-driven industrial process modeling methods are currently widely used [1].

There are usually many defects in the original process data, such as inconsistency, missing, noise, or redundancy. So the data preparation is one of the important stages of the data-driven modeling process. The purpose of the preparation phase is to eliminate the defects mentioned above and provide high-quality data for subsequent learning algorithms [2]. As the amount of process data increases, the data stream becomes a common analysis object. In the practical industrial process, the process variables are obtained by using the PLC/DCS system with a continuous data stream style. The non-stationarity of the data stream with time changes places higher requirements for data preparation. This feature has attracted the attention of many scholars. Especially, when the process data distribution changes over time, the old data-driven model is gradually not suitable for current data. This phenomenon is also called concept drift [3].

Regarding the issue above, online learning methods are widely adopted, because their decision-making mechanism will be updated in real-time to deal with concept drift [4]. In recent years, for different learning tasks and learning scenarios, a large number of online learning algorithms have been produced. Ref [5] proposed a perceptron algorithm implemented with an online learning model. It was generalized as a non-linear perceptron method based on kernel functions [6]. The online passive active algorithms [7], LASSO online learning algorithms [8], regularized dual average online learning methods [9], et al. are also proposed. However, for the concept drift problem, online learning algorithms still face various challenges, such as: (1) how to ensure the stability of online learning updates when the distribution of new and old modeling samples is inconsistent; (2) how to effectively learn when the sample concepts are different and are fused into the new model [10]; and (3) how to identify drift samples when the labeled samples are sparsely acquired.

In response to the above challenges, research on concept drift detection has developed rapidly in recent years. Fig. 1 shows the growth of refs and citations related to concept drift in the past 20 years [11]. As we know, the earlier drift detection system is the FLORA algorithm series [12]. The algorithm initially realizes the distinction and storage of sample concepts. In subsequent work, the causes, types, and definitions of concept drift have been further improved [13]. Different learning methods for concept drift such as semi-supervised learning, active learning, and repetitive concept learning have been proposed successively [14, 15, 16]. Moreover, the existing model is also combined with drift detection. It is constructed as a specific drift adaptation model [17]. In industrial data-driven models, concept drift has three main detection methods, i.e., based on process

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variables, based on prediction error of difficulty-to-measure parameter, and based on combine multiple factors. It can also be grouped as single sample and multi-sample detection methods.

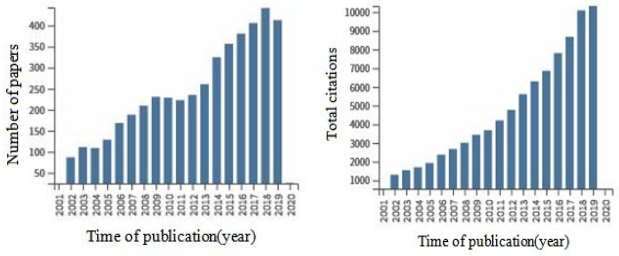


Fig. 1: Research trends in concept drift

At present, there are many review studies about concept drift [18 , 19]. They comprehensively introduced the research progress. Among them, ref [18] summarized the drift detection algorithm based on classification tasks, and ref [19] summarized it from three perspectives, i.e., drift detection, understanding, and adaptation. They are not address on the concept drift problem on industrial process modeling.

Therefore, compared with other review works, this paper focuses on the background of industrial processes modeling. The main research focuses are: (1) enumerating commonly used concept drift detection methods in the industry; (2) summarizing the current research results of drift detection, discuss methods based on the above three detection methods; (3) provide insights into problems and development directions in this field.

2 Description of Concept Drift for Industrial Processes

2.1 General Definition and Classification of Concept Drift

Concept drift refers to the statistical characteristics of the target sample change in a random manner over time [20]. This description was first proposed by ref [21], whose purpose is to point out that noisy data will be transformed into non-noisy data in some cases. Moreover, these changes are usually difficult to measure [22].

The general form of concept drift in industrial processes is as follows. Given the set of modeling samples at time [0, t], $d_i = (\mathbf{X}_i, y_i)(i \in [0,t])$ is a sample in $S_{0,t} = \{d_0, \ldots, d_t\}$, where $\mathbf{X}_i$ is the model input (this paper refers to the process variables), y_i is the model output truth value (this paper refers to the difficulty-to-measure parameter), and within $S_{0,t}$ the samples all follow the distribution $F_{0,t}(\mathbf{X}, y)$. It is assumed that the new sample $d_k (k \in [t+1, \infty))$ follows the distribution $F_k(\mathbf{X}, y)$. When $F_{0,t}(\mathbf{X}, y) \neq F_k(\mathbf{X}, y)$, the concept drift of the sample d_k is considered.

According to different research priorities, researchers divide concept drift into different categories. For example, ref [23] divided it into virtual and real concept drifts according to different data generation environments; ref [24] proposed a concept drift generation model based on the multi-source of data generation. At present, the most commonly used classification methods are described as four types, which are sudden drift, gradual drift, incremental drift, and reoccurring drift [3]. Examples are shown in Fig. 2:

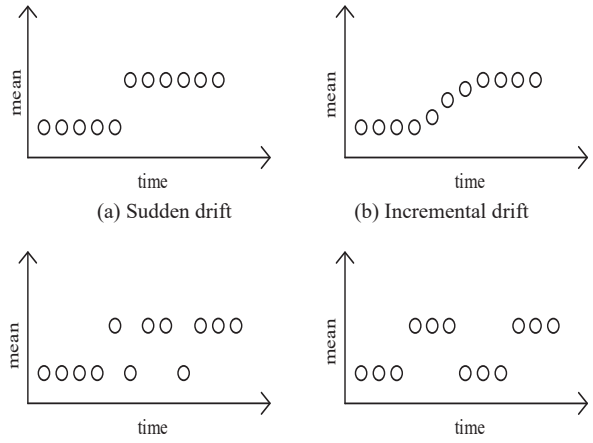


Fig. 2: Illustration of the four drift types

2.2 Causes and Classification of Industrial Process Concept Drift

Common types of concept drift in industrial processes are sudden drift and gradual drift, of which reoccurring drifts are rare due to the complex changes in the industrial environment. According to two common causes of drift in industry, it is divided into process drift and sensor drift [25].

There are two general reasons for the process drift. The first one is the change of the process itself, such as the stable wear of mechanical components in the factory during operation. The second one is the change in the external conditions of the process, such as weather changes and catalyst deactivation. In the municipal solid waste incineration process, the component and water content of the waste will change according to the living level and seasons.

Sensor drift, also known as measurement drift, is caused by changes in feedback from the measurement device. Some researchers believe that it does not reflect changes in the process itself and is therefore not considered drift. But in a data-driven model, any change in the data may affect the decision and update of the model. Therefore, this kind of drift should be distinguished in time, and the input and output changes of the sensor should be discriminated against.

There are many concept drift detection methods that have been incorporated into industrial processes. Ref [26] applied detection methods based on non-parametric U test, local performance parameter test and statistical analysis of raw data to online prediction of mass flow in circulating fluidized bed boilers. Model prediction error is used to detect drift in the research of industrial radial fan adaptive maintenance, [27]. Ensemble regression model is used to detect drift during the etching process of semiconductor manufacturing [28].

2.3 Process of Concept Drift

Ref [19] proposed a general concept drift processing flow, as shown in Fig. 3.

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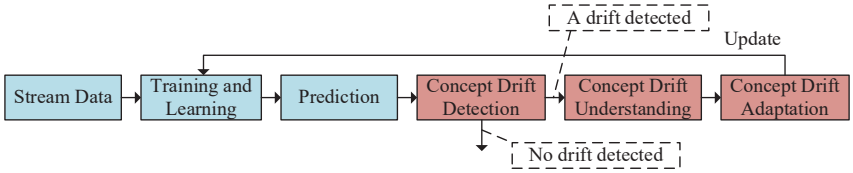


Fig. 3: Process of concept drift

In ref [19], the process of concept drift is divided into three steps, i.e., detection, understanding, and adaptation. It also pointed out that some methods can perform drift detection and understanding at the same time. Drift detection refers to the technology and mechanism of characterizing and quantifying concept drift by identifying change points or change intervals. Drift understanding solves the problems of "when", "where" and "how", that is, to obtain drift obtaining state information such as time, area, and degree, and using it as input for drift adaptation. Drift adaptation usually uses the method of simple retraining, comprehensive retraining, or model adjustment to update the model based on the drift state information. It should be noted that the prediction error cannot be obtained in most of the actual industrial processes because the true value of the difficulty-to-measure process is difficult to obtain. For example, in the process of municipal solid waste incineration, the concentration of pollutants called dioxin is difficult to be detected online in real-time. Thus, process variables-based methods have to be used in this case.

3 Drift Detection Methods Review

This section divides the drift detection method into three categories, i.e., based on process variables, based on prediction error of difficulty-to-measure parameter, and based on combine multiple factors.

3.1 Methods Based on Process Variables

The process monitoring method based on principal component analysis (PCA) has been successfully applied in industrial processes with time-varying characteristics [29]. At first, the method divided the new sample d_{t+1} into two parts, i.e., $\hat{d}_{t+1}$ and $\tilde{d}_{t+1}$, which are projections of d_{t+1} on the principal subspace and residual subspace of the PCA model, respectively. Then, the squared prediction error (SPE) and Hoteling's T^2 index of the new sample are calculated [30]. Generally, *SPE* is used to measure the projection of the new sample on the residual subspace. It indicates the degree of deviation of the new sample from the model. Moreover, T^2 is used to measure the change of the new sample on the principal component subspace. It indicates the degree of deviation of the new sample within the model. Finally, it is judged that the new sample concept does not drift when *SPE* and T^2 meet the control limits of *SPE* and T^2. In the recent researches, a weighted recursive PCA detection algorithm was proposed [31]. It can effectively reduce the computational complexity. Fuuther, a detection method combining exponentially weighted moving average (EWMA) and PCA was used for fault detection of cement plants [32]. Recently, a method based on signed directed graph (SDG) and PCA algorithms are used in online process monitoring systems [33].

The approximate linear dependence (ALD) value between the old training samples and the new sample is also used to detect the sudden and progressive drift [34, 35]. The ALD value is calculated in the input sample's original space or kernel space firstly. Then, the drifting is judged according to the relationship between δ_{t+1} and threshold v. That is to say, when $\delta_{t+1} \le v$, it is considered that the new sample not drift. Otherwise, when $\delta_{t+1} > v$, it is considered that the new sample drift. The threshold v is determined according to the requirements of modeling accuracy and speed in the actual industrial system. In nature, it can usually be expressed as a single-objective optimization problem [36].

3.2 Methods Based on Difficulty-to-measure Parameter

Ref [37] proposed a detection algorithm based on the prediction error band (PEB). When the PEB meets the conditions $e_k \le Rule\left(\left\{\delta_k^1, \delta_k^2, \cdots\right\}\right)$, the sample is considered not drift. In which, $\delta_k^1, \delta_k^2, \cdots$ represents different thresholds set based on prior knowledge and $Rule(\cdot)$ represents judgment rules set based on experience. The representative algorithm based on model's prediction error is DDM [38]. It is realized with three steps: (1) define the sample warning level and drift level; (2) use a window to collect new samples; and (3) calculate the prediction error on the new samples. When the error change amplitude reaches the warning level, it is ready to build a new model. Moreover, the old model is still used to predict at this time. However, when the error change amplitude reaches the drift level, the old model will be replaced by a new one. The training samples are obtained accumulated during the process from warning level to drift level. That is to say, DDM provides a preliminary detection framework, which is to complete the collection, detection, judgment, and update of new samples through the window, prediction error, and level definition.

The above methods are based on single samples. In the detection method for multiple samples, the distance function is generally used to quantify the distribution difference between the old samples and new ones. When the sample distance is proved to be statistically significant, a concept drift alarm will be issued. Then, the old model will enter the update process.

Ref [39] earlier provided guidance for the design of distance function in multi-sample distribution difference analysis. Then, it proposed relative distance families. The significance level of the distance family was determined by Chernoff bound and Vapnik Chervonenkis dimension. In another method, each sample block is mapped to a concept vector after the historical samples are divided into multiple sample blocks. Further, multiple concept vectors are clustered into a concept. When a new sample block arrives, its concept vector and each cluster center distance is used to detect drift [40]. The above algorithm process can be represented by Fig. 4. It can also represent a general multi-sample detection process.

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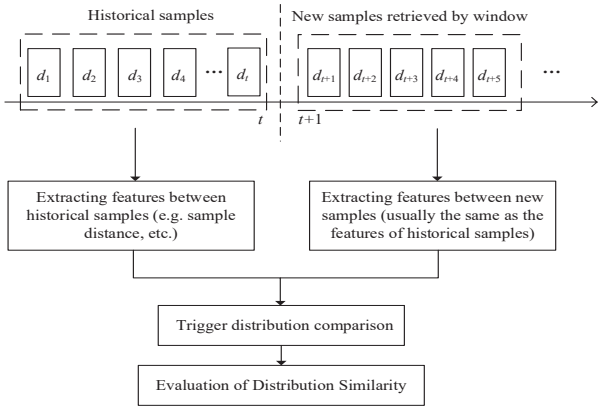


Fig. 4: Multi-sample drift detection process

Similarly, there is a method by using the central limit theorem without relying on the sample distribution [41]; and a method used the entropy value to calculate the difference in the sample block distribution [42]. In recent research, a concept neuron algorithm proposed by ref [43]. It adopts a window to monitor the residual distribution in the new sample block and to update the model with a parameter inverse gradient based on the residual after monitoring the change. Ref [44] proposed a change detection experiment based on the least square density difference.

3.3 Method Based on Combine Multiple Factors

Ref [45] combined ALD and relative prediction error (RPE) methods by using fuzzy rules to judge concept drift. Compared with only ALD-based algorithms and only PEB-based methods, this approach combined the model's predictive performance change and the drift characteristic of process variable. Moreover, it can fuse domain expert knowledge based on the actual industrial process. However, the method was not verified with real data, and the statistical test of sample distribution didn't consider.

Ref [46] combined PCA and the statistical method based on the Wilcoxon test to monitor change of the sample class distribution and the posterior probability distribution. At the same time, a data structure named P-tree is constructed to store samples of concept changes to achieve the purpose of efficient online model maintenance. The framework of this algorithm is shown in Fig. 5.

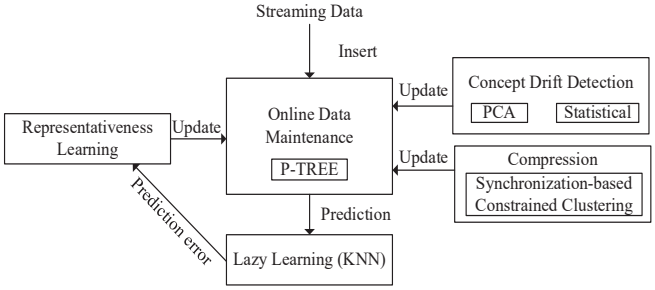


Fig. 5: Multi-scale model maintenance framework

This framework can also be used as a general multi-scale model maintenance framework.

4 Discussion and Analysis

4.1 Methods Comparison

The characteristics of the above detection methods are compared here, and the results are shown in Table 1.

Table 1 Comparison of Concept Drift Detection Methods

number	Ref number	Year	Research content	Method category	Advantage	Disadvantage
1	[43]	2016	A method based on concept neuron	Method based on difficulty-to-measure parameter	Fast detection speed, can find abnormalities in time	Lack of representation of drift state information
2	[44]	2016	A method based on least-squares density difference			
3	[45]	2016	A method of combining ALD and RPE	Method based on combine multiple factors	Can balance detection speed and performance	Reliance on specific scenario modeling, it is difficult to establish standardized indicators
4	[46]	2018	A method of combining statistical-based and PCA			
5	[33]	2018	A method of combining SDG and PCA	Method based on process variables	Can accurately reflect sample changes	Difficult to balance model accuracy and update speed, lack reasonable trigger mechanism
6	[27]	2019	An adaptive model maintenance based on prediction error	Methods based on difficulty-to-measure parameter	Fast detection speed, can find abnormalities in time	Lack of representation of drift state information

According to Table 1, the overall detection speed of the detection method based on difficulty-to-measure parameter relatively fast. It can reflect the possible time and location of the drift for the model in time. Thus, the model can take measures to deal with the concept drift that has just started or is about to start. However, it is difficult for this detection method to reflect more drift state information and to update the model in a targeted manner.

The detection based on process variables is usually implemented by a distribution-based method. Thus, the accuracy of the test results is guaranteed. Moreover, it can usually reflect a certain degree of drift and distribution difference characteristics. Therefore, the model can be selectively updated efficiently. However, the retrieval and inspection of samples in this method are usually slower and consume more computing resources.

The method based on combine multiple factors makes the method have the advantages of the above two methods. But this requires that the connection between the two methods is reasonable and has a scientific distribution of computing

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power. At the same time, the method faces the same problem as the method based on difficulty-to-measure parameter, that is, it cannot be used when the true value is difficult to obtain.

4.2 Difficulties Analysis

Through the above analysis, it is found that the following issues still need to be resolved in the field of concept drift detection for industrial processes.

(1) How to obtain the sample expected distribution of the actual industrial process

Although the research on concept drift detection has achieved fruitful results, there is still no strict mathematical description of the definition, speed, and amplitude of the concept. Therefore, when detecting the condition of sample drift, the differences in sample distribution are usually reflected indirectly through indicators such as prediction error or sample distance. Thus, it is difficult to obtain accurate representations of the sample concept and distributions. At the same time, it is difficult to obtain some real values in the actual industrial process. Moreover, it is difficult to apply the supervised detection method that does not rely on the sample distribution. Therefore, future work should strengthen the study of sample distribution expression and combine it with unsupervised or semi-supervised learning methods. That is to say, the corresponding methods based only on process variables or based on a small number of truth values.

(2) How to mine more information from methods based on difficulty-to-measure parameter.

At present, most methods based on difficulty-to-measure parameter are limited to detecting samples as abnormal samples. However, they have not thoroughly studied other drift information carried by the samples, such as the time when the drift may occur or the possibility of imminent drift. For this reason, as far as possible, a quantitative description of the abnormality of each sample should be described. Thus, the model can be remind that how to prepare the next step to ensure rapid adaptation when the drift arrives.

(3) How to make process variables based method more accurate.

To obtain higher prediction accuracy and update speed in actual industrial models, selective updating and forgetting should be performed according to the degree or type of drift of samples. Thus, this requires that the process variables can accurately express the difference between its distribution information and historical sample distribution. Current research focuses on existence, not the extent.

5 Conclusions and Prospect

This paper first introduced the general processes and challenges of current data-driven modeling in the context of industrial processes. Then, it summarized the definition of concept drift, research status, and industrial applications. Finally, analyzed some problems that need to be solved in this field at this stage.

Further research is needed in the following aspects:

(1) Research on define sample distribution expression. The distance function can be combined with a variety of statistical testing methods. It can be used to analyze the correlation degree and change characteristics of each process variables. Thus, an accurate sample distribution measurement method need to be established.

(2) Research on improvement of method based on prediction error of difficulty-to-measure parameter. After determining the sample distribution expression, the detection algorithm can further combine with PCA, PEB, sample distance or other indicators of change. Then, the new drift threshold indicators should be define to determine the degree of drift.

(3) Research on improvement of process variables based method. The sample retrieval mechanism is determined according to the sampling rate and the number of samples. The distribution characteristics of the sample block are summarized by clustering and other methods to establish a drift index. When a new sample block is detected drift, the model update method can be guided according to the similarity of the new samples and modeling samples.

References

[1] E. R. Xin, Z. Dong, and X. L. Cao, Dynamic data driven modeling of industrial system based on state optimization, *Computer Simulation*, 35(5): 139-142, 2018.

[2] S. García, L. Julián, and H. Francisco, Data preprocessing in data mining, *Springer International Publishing*, 2015.

[3] I. Žliobaitė, Learning under concept drift: an overview, *Computer Science*, 2010.

[4] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, A survey on concept drift adaptation, *ACM computing surveys*, 46(4): 1-37, 2014.

[5] F. Rosenblatt, The perceptron: a probabilistic model for information storage and organization in the brain, *Psychological review*, 65(6): 386-408, 1958.

[6] S. S. Shwartz, Online learning and online convex optimization, *Foundations and Trends in Machine Learning*, 4(2): 107-194, 2012.

[7] K. Crammer, O. Dekel, J. Keshet, S. S. Shwartz, and Y. Singer, Online passive-aggressive algorithms, *Journal of Machine Learning Research*, 7(Mar): 551-585, 2006.

[8] L. Z. Jie, L. Y. Xiang, W. Feng, F. Yu, and Z. L. Xiang, Efficient and accelerated online learning for sparse group LASSO, in: *2014 IEEE International Conference on Data Mining Workshop*, 2014: 1171-1177.

[9] L. Xiao, Dual averaging methods for regularized stochastic learning and online optimization, *Journal of Machine Learning Research*, 11(Oct): 2543-2596, 2010.

[10] Z. J. Li, Y. X. Li, F. Wang, G. L. He, and L. Kuang, Online learning algorithms for big data analytics: a survey, *Journal of Computer Research and Development*, 52(8): 1707-1721, 2015.

[11] Citation report of concept drift, *Web of Science,* www.isiknowledge.com.

[12] G. Widmer and M. Kubat, Learning in the presence of concept drift and hidden contexts, *Machine Learning*, 23(1): 69-101, 1996

[13] R. ELWELL and R. POLIKAR, Incremental learning of concept drift in nonstationary environments, *IEEE Transactions on Neural Networks*, 22(10): 1517-1531, 2011.

[14] J. GAO, B. DING, W. FAN, J. Han, and S. Y. Philip, Classifying data stream with skewed class distributions and concept drifts, *IEEE Transactions on Internet Computing*, 12(6):37-49, 2008.

[15] J. C. Xue and G. M. Weiss, Quantification and semi-supervised classification methods for handling changes in class distribution, in: *Proceedings of the 15th ACM*

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SIGKDD international conference on Knowledge discovery and data mining, 2009: 897-906.
[16] P. LI, X. D. WU, and X. G. HU, Learning from concept drift data streams with unlabeled data, in: *Proceedings of the 24th AAAI Conference on Artificial Intelligence*, 2010: 1945-1946.
[17] Y. M . WEN, Y. N. WANG, and Y. ZHANG, Fast detection of concept drifts based on confident majority voting, *Journal of Hunan University: Natural Sciences*, 37(6): 36-40, 2010.
[18] Y. M. Wen, B. H. Qiang, and Z. G. Fan, A survey of the classification of data streams with concept drift, *CAAI Transactions on Intelligent Systems*, 8(2): 95-104, 2013.
[19] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Q. Zhang, Learning under concept drift: a review, *IEEE Transactions on Knowledge and Data Engineering*, 31(12): 2346-2363, 2018.
[20] N. Lu, G. Zhang, and J. Lu, Concept drift detection via competence models, *Artificial Intelligence*, 209: 11-28, 2014.
[21] J. C. Schlimmer and R. H. Granger, Incremental learning from noisy data, *Machine learning*, 1(3): 317-354, 1986.
[22] A. Liu, Y. Song, G. Zhang and J. Lu, Regional concept drift detection and density synchronized drift adaptation, in: *Proc. 26th Int. Joint Conf. Artificial Intelligence*, 2017: 2280–2286.
[23] G. Widmer and M. Kubat, Effective learning in dynamic environments by explicit context tracking, in: *European Conference on Machine Learning*, 1993: 227-243.
[24] A. M. Narasimhamurthy and L. I. Kuncheva, A framework for generating data to simulate changing environments, in: *Artificial Intelligence and Applications*, 2007: 415-420.
[25] P. Kadlec, B. Gabrys, and S. Strandt, Data-driven soft sensors in the process industry, *Computers & chemical engineering*, 33(4): 795-814, 2009.
[26] J. Bakker, M. Pechenizkiy, I. Žliobaitė, A. Ivannikov, and T. Kärkkäinen, Handling outliers and concept drift in online mass flow prediction in CFB boilers, in: *Proceedings of the Third International Workshop on Knowledge Discovery from Sensor Data*, 2009: 13-22.
[27] J. Zenisek, F. Holzinger, and M. Affenzeller, Machine learning based concept drift detection for predictive maintenance, *Computers & Industrial Engineering*, 137: 106031, 2019.
[28] F. Rosenthal, P. B. Volk, M. Hahmann, D. Habich, and W. Lehner, Drift-aware ensemble regression, in: *International Workshop on Machine Learning and Data Mining in Pattern Recognition*, 2009: 221-235.
[29] J. Tang, T. Y. Chai, Z. Liu, and X. J. Zhou, Adaptive ensemble modelling approach based on updating sample intelligent identification, *Acta Automatica Sinica*, 42(7): 1040-1052, 2016.
[30] H. H. Yue, and S. J. Qin, Reconstruction-based fault identification using a combined index, *Industrial & engineering chemistry research*, 40(20): 4403-4414, 2001.
[31] I. Portnoy, K. Melendez, H. Pinzon, M. Sanjuan, An improved weighted recursive PCA algorithm for adaptive fault detection, *Control Engineering Practice*, 50: 69-83, 2016.
[32] A. Bakdi, A. Kouadri, and A. Bensmail, Fault detection and diagnosis in a cement rotary kiln using PCA with EWMA-based adaptive threshold monitoring scheme, *Control Engineering Practice*, 66: 64-75, 2017.
[33] X. Han, S. Tian, J. A. Romagnoli, H. Li, and W. Sun, PCA-SDG based process monitoring and fault diagnosis: application to an industrial pyrolysis furnace, *IFAC-PapersOnLine*, 51(18): 482-487, 2018.
[34] Y. Engel, S. Mannor, and R. Meir. The kernel recursive least-squares algorithm, *IEEE Transactions on signal processing*, 52(8): 2275-2285, 2004.
[35] J. Tang, W. Yu, T. Y. Chai, and L. Zhao, On-line principal component analysis with application to process modeling, *Neurocomputing*, 82: 167-178, 2012.
[36] J. Tang, T. Y. Chai, and L. J. Zhao. On-line KPLS algorithm with application to ensemble modeling parameters of mill load, *Acta Automatica Sinica*, 39(5): 471-486, 2013.
[37] Y. Liu, H. Wang, J. Yu, and P. Li, Selective recursive kernel learning for online identification of nonlinear systems with NARX form, *Journal of Process Control*, 20(2): 181-194, 2010.
[38] J. Gama, P. Medas, G. Castillo, and P. Rodrigues, Learning with drift detection, in: *Proc. 17th Brazilian Symp. Artificial Intelligence*, 2004: 286–295.
[39] D. Kifer, S. B. David, and J. Gehrke, Detecting change in data streams, in: *Proc. 30th Int. Conf. Very Large Databases*, 2004: 180–191.
[40] I. Katakis, G. Tsoumakas, and I. Vlahavas. Tracking recurring contexts using ensemble classifiers: an application to email filtering, *Knowledge and Information Systems*, 22(3): 371-391, 2010.
[41] C. Alippi and M. Roveri. Just-in-time adaptive classifiers—Part I: Detecting nonstationary changes, *IEEE Transactions on Neural Networks*, 19(7): 1145-1153, 2008.
[42] P. Vorburger, A. Bernstein. Entropy-based concept shift detection, in: *Sixth International Conference on Data Mining*, 2006: 1113-1118.
[43] L. Moreira-Matias, J. Gama, and J. M. Moreira, Concept neurons–handling drift issues for real-time industrial data mining, in: *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, 2016: 96-111.
[44] L. Bu, C. Alippi, and D. Zhao, A pdf-free change detection test based on density difference estimation, *IEEE transactions on neural networks and learning systems*, 29(2): 324-334, 2016.
[45] J. Tang, T. Y. Chai, Z. Liu, and X. J. Zhou, Adaptive ensemble modelling approach based on updating sample intelligent identification, *Acta Automatica Sinica*, 42(7), 1040-1052, 2016.
[46] J. Shao, F. Huang, Q. Yang, and G. Luo, Robust prototype-based learning on data streams, *IEEE Transactions on Knowledge and Data Engineering*, 30(5): 978-991, 2017.

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